

ISOBARIC VAPOR-LIQUID EQUILIBRIUM FOR THE BINARY MIXTURE OF ETHANOL (1) + 1-HEXANOL (2) AT 100 kPa

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ABSTRACT

The isobaric vapor-liquid equilibrium (VLE) data for the binary mixture of ethanol + 1-hexanol were determined at 100 kPa through the measurements of boiling point temperature by using modified Othmer recirculation still. No azeotrope was formed in the investigated binary system. The Wilson, nonrandom two-liquid (NRTL), and universal quasichemical (UNIQUAC) models were used to correlate the experimental data to obtain binary interaction parameters. These three models yield satisfactory results.

Keywords : ethanol, 1-hexanol, vapor-liquid equilibrium

Introduction

In the recent years, the oil price has been rising globally. On the other hand, the number of new oil resources are significantly decrease. Alcohol is one of the promising alternative energy to overcome these problem. Mixed alcohols which consists of C1-C8 alcohols can be produced by using coal through new chemical synthesis technology (Wang & Bao, 2013). Distillation process is employed to separate those mixed alcohols in order to obtain single alcohol such as ethanol, 2-butanol, 1-pentanol, 1-hexanol, etc. To develop the separation process, vapor-liquid equilibrium (VLE) data of the related mixtures are essential and play important rule in the design and optimization of the process.

Several studies on the VLE measurement of mixed alcohols have been published. The isobaric VLE data for binary systems of methanol with 1-propanol, 1-butanol, and 1-pentanol at 101.3 kPa were reported by Hill & Van Winkle (1952). For the system containing ethanol, isobaric VLE data of ethanol with 1-butanol, 2-butanol, and 1-pentanol at 101.3 kPa were reported by Hellwig & Van Winkle (1953), with isobutanol at 101.3 kPa by Resa et al.

(2004), with tert-butanol at 101.3 kPa by Yang & Wang (2002). For 1-propanol system, Ballard & Van Winkle (1952) reported the isobaric VLE of 1-propanol with 2-propanol at 101.3 kPa. For the binary mixture containing 2-propanol, isobaric VLE data of 2-propanol with methanol, ethanol, and 2-butanol at 101.3 kPa were determined by Ballard & Van Winkle (1952), with 1-butanol and 1-pentanol at 101.3 kPa by Wang & Bao (2013), with 2-butanol at 101.3 kPa by Tamir & Wisniak (1975). The isobaric VLE data at 101.3 kPa of 1-butanol with 2-butanol and tert-butanol were reported by Wisniak & Tamir (1976), with isobutanol at 101.3 kPa by Tamir & Wisniak (1975), with 1-propanol at (53.3 and 91.3) kPa by Mohsen-Nia & Memarzadeh (2010). The other binary mixture, isobutanol + tert-butanol was measured at 94.9 kPa by Darwish & Al-Anber (1997). The systems containing binary mixture of ethanol + 1-hexanol has not been fully investigated.

In this present study, the isobaric VLE data were measured through boiling point for the binary systems of ethanol with 1-hexanol at 100 kPa by using an Othmer-type recirculation still. The VLE data of ethanol + 1-hexanol measured in this study have not

been found in the investigated pressure. The experimental data were correlated using the activity coefficients models such as the Wilson (Wilson 1964), NRTL (Renon & Prausnitz 1968), and UNIQUAC (Abrams & Prausnitz 1975).

Research Methods

Materials

The specification of materials used in this study were listed in Table 1. Ethanol was used without additional purification. 1-Hexanol was dried using 3A molecular sieve adsorption with final mass fraction purity better than 0.989.

Table 1. The Specification of Materials

Compound	Source	Mass Fraction Purity	MW (g·mol ⁻¹)
Ethanol	Merck	0.999	46.069
1-Hexanol	Merck	0.989	102.177

Apparatus and procedures

The boiling temperatures were measured using an Othmer-type recirculation still. The detailed diagram and working procedure of the apparatus were explained by Morrison et al. (1990). The top part of the equilibrium still condensor was opened to the air to ensure the pressure inside the still in standard atmospheric condition. The pressure condition of the experimental environment was 100±0.2 kPa. The atmospheric pressure was measured using calibrated-TFA barometer (Germany) which has stability ±0.1 kPa. The mixture was injected about 150 cm³ in each experimental run. The temperature of the mixtures was kept for at least 1 hour to ensure equilibrium conditions achieved. The equilibrium temperature was measured by using a calibrated K type-digital thermometer (TK4S-14RN, USA) with uncertainty ±0.1 K. As equilibrium temperature was attained, liquids in equilibrium condition were collected for analysis. A gas chromatograph (GC) (Agilent 6820, USA) equipped with

flame ionization detector (FID), HP-5 Column (P/N : 19091J-413) and a carrier gas of helium with purity >0.9995 in mass fraction was used to analyze the liquid phase composition. The temperature operations of GC were 384.15 K for the front inlet, 573.15 K for the column, and 523.15 K for the detector temperature.

Results and Discussion

The boiling temperature data have been measured for the binary system of ethanol (1) + 1-hexanol (2) at 100 kPa as listed in Table 2 and the *T*-*x*-*y* diagram was shown in Figure 1.

Table 2. Vapor-Liquid Equilibrium Data for Binary System of Ethanol (1) + 1-Hexanol (2) at 100 kPa^a

<i>x</i> ₁	<i>T</i> (K)
0.0000	429.2
0.0195	419.0
0.1499	395.7
0.2045	391.1
0.3128	376.5
0.3778	374.3
0.4647	368.5
0.5448	365.0
0.6179	364.4
0.6855	359.8
0.7662	357.9
0.8273	354.4
0.8649	352.5
0.9157	351.7
1.0000	351.6

^a *u*(*x*₁) = 0.001, and *u*(*T*) = 0.1 K

The experimental VLE data were correlated using the Wilson, NRTL, and UNIQUAC models. The relationship of vapor and liquid phases in equilibrium is described as follows:

$$y_i \Phi_i P = x_i \gamma_i P_i^{sat} \quad (1)$$

where *y*_{*i*}, *x*_{*i*}, Φ_i , γ_i , and *P* refer to vapor phase composition, liquid phase composition,

vapor phase correction factor, activity coefficient and pressure, respectively. The superscript *sat* stands for saturated, and the subscript *i* represent component *i*. The vapor phase correction factor in this work was considered 1 as represent the ideal vapor phase.

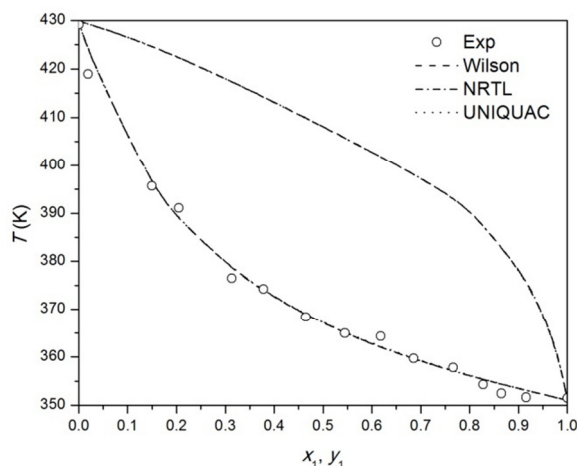


Figure 1. VLE phase (T - x_1 - y_1) diagram for binary system of ethanol (1) + 1-hexanol (2) at 100 kPa

The objective function used for the optimization is shown in equation 2.

$$OF = \sum_k^N (T_k^{\text{exp}} - T_k^{\text{cal}}) \quad (2)$$

where N and k are the total number of experimental data points and the point, respectively. T_k^{exp} and T_k^{cal} refer to experimental temperature and calculated temperature in equilibrium, respectively.

Antoine equation was used to calculate the saturated pressure of pure component as shown in equation 3.

$$\log P^{\text{sat}} = A - \frac{B}{T + C - 273.15} \quad (3)$$

where P^{sat} is in bar and T is in K. A , B , and C are the parameters of the Antoine

equation for each component as listed in Table 3 (Poling et al., 2001).

Table 3. Parameters of the Antoine Equation for Pure Compounds

Component	A	B	C
Ethanol	5.33675	1648.22	230.918
1-Hexanol	4.18948	1295.59	152.510

The physical properties and parameters of each component used in the activity coefficients correlation for the Wilson, NRTL, and UNIQUAC models are given in Table 4.

Table 4. Physical Properties and Parameters of Pure Components Used in the Activity Coefficients Correlation

Component	V^a ($\text{cm}^3 \cdot \text{mol}^{-1}$)	r^b	q^b
Ethanol	58.68	2.5755	2.588
1-Hexanol	125.19	5.2731	4.748

^a Poling et al., 2001

^b Hansen et al., 1991

The activity coefficients models used in this study are able to correlate well to the experimental data. As shown in Figure 1, the deviations between the experimental and calculated points are small. The best fitted binary interaction parameters and the average absolute deviations (AAD) for each model were listed in Table 5.

Table 5. Fitted Binary Interaction Parameters of Activity Coefficient Models and Average Absolute Deviations (AAD) for Binary System of Ethanol (1) + 1-Hexanol (2) at 100 kPa.

Model	Parameters		AAD ^a (%)
	A_{12} (K)	A_{21} (K)	
Wilson ^b	-91.214	-45.200	0.32
NRTL ^c	513.665	-331.956	0.32
UNIQUAC ^d	-11.287	-17.367	0.32

^a $AAD = (1/n) \sum_{i=1}^n |(T_{\text{cal}} - T_{\text{exp}})| / T_{\text{exp}} \cdot 100\%$, where n is the number of data points.

^b $\Lambda_{ij} = (V_j/V_i) \exp(A_{ij}/T)$, where V is the molar volume of the component.

^c $\tau_{ij} = A_{ij}/T$, the value of α was fixed to be 0.3.

^d $\tau_{ij} = \exp(A_{ij}/T)$

Conclusions

The isobaric VLE data have been measured experimentally for the binary system of ethanol (1) + 1-hexanol (2) at 100 kPa by using modified Othmer recirculation still through boiling points temperatures. The binary system investigated in this study has no azeotrope formation. The Wilson, NRTL, and UNIQUAC models were used to correlate the VLE data of binary system. These models showed satisfactory results in the VLE data correlation.

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